

Atomic Models

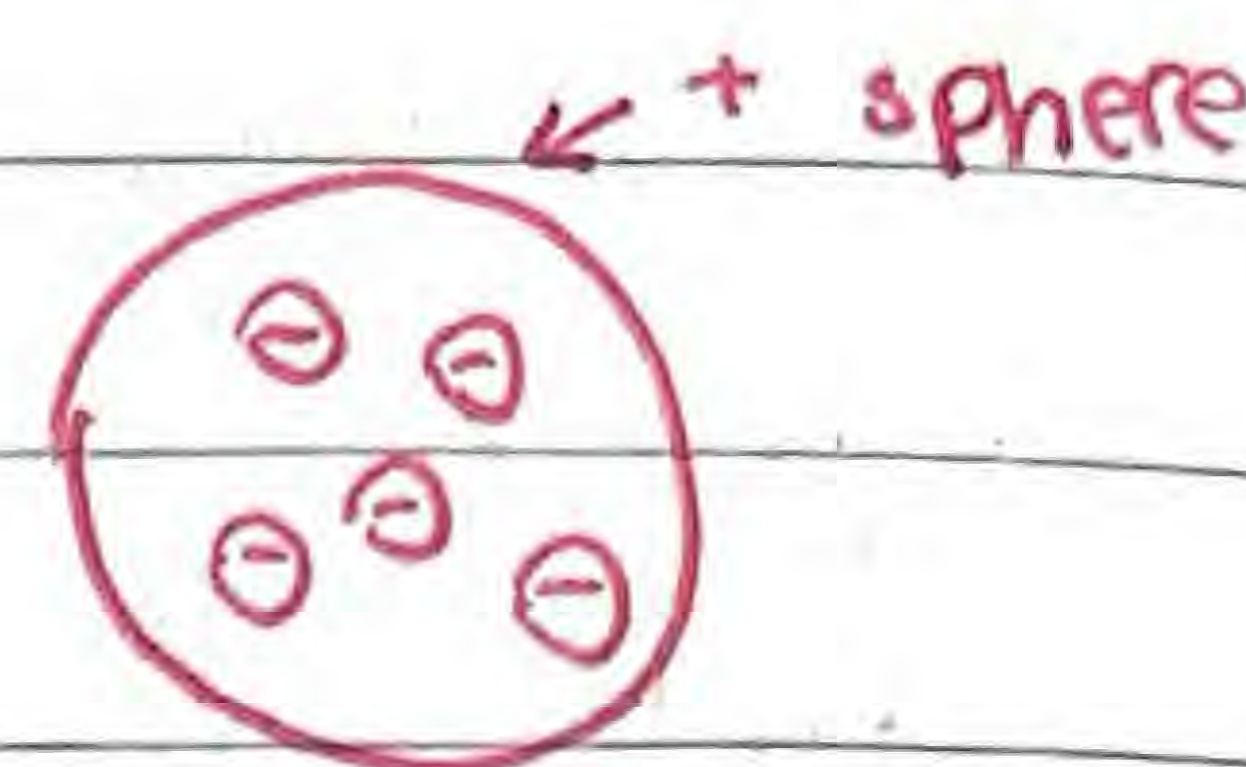
• Thomson Model (1897)

> "Pie"-like model

↳ plum like electrons

↳ pudding like sphere

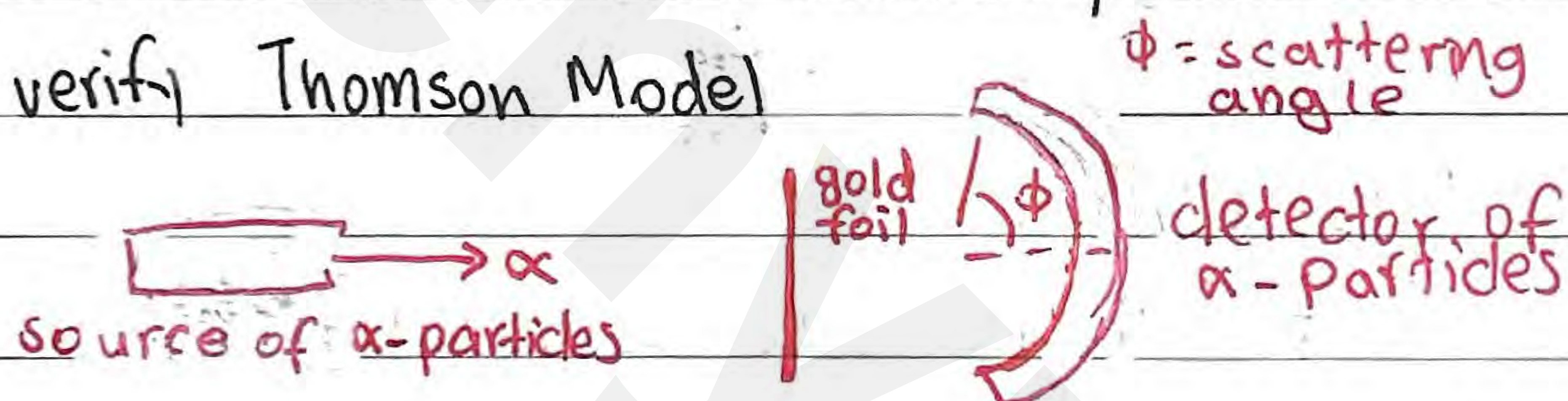
> No experiments done to verify (purely theoretical)



• Rutherford Experiment (1909)

> Also known as Geiger-Marsden, Gold foil experiment

> To verify Thomson Model



* α are used because

• Ability to ionize (γ can't)

• Highest ionization power (largest charge)

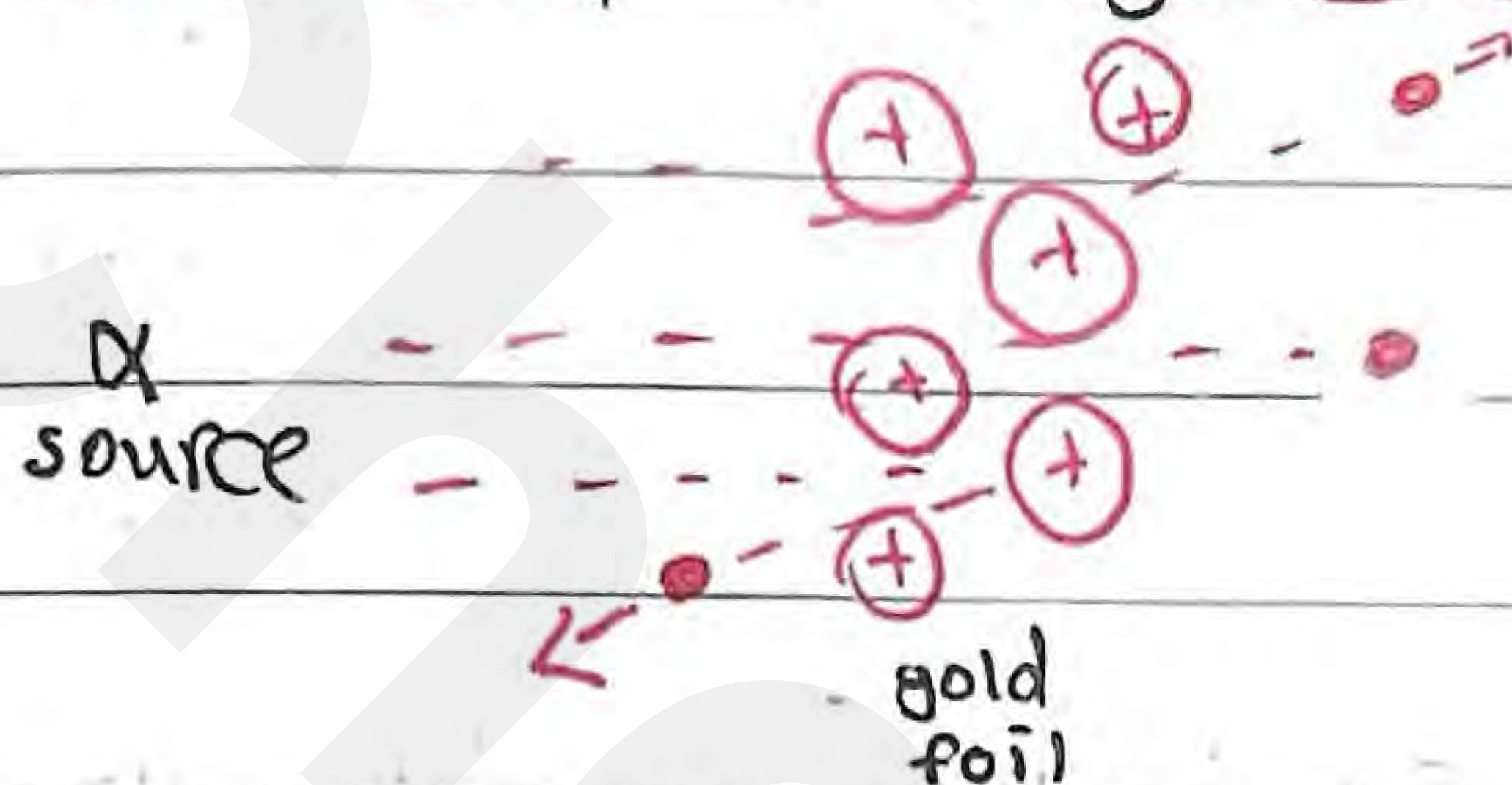
• Large size (to match gold foil)

Gold foil is used because of high malleability (so a thin layer can be created, easier penetration)

If Thomson model were correct then uniform charge distribution means no scattering angle

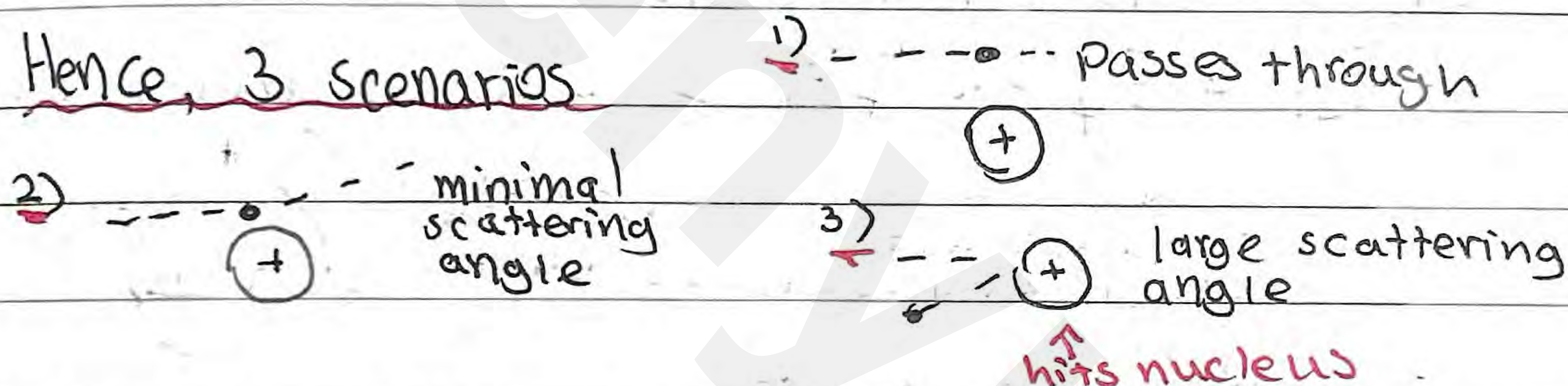
Rutherford Experiment

But Rutherford found that while the great majority went straight through the foil, a lot were scattered at small angles, very occasionally at larger angles (deflection)



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Hence, 3 scenarios



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This revealed that nucleus is small in comparison to entire atom

↳ Atom is mostly empty space

↳ Nucleus is $\approx 10^{-15}$ m, Atom is $\approx 10^{-10}$ m

yet they appear "solid" because of repulsive electrostatic forces between objects

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Consider...

decelerates until $v=0$, then repulsed



distance by which it's deflected

if α has lower E_k , more easily deflected. Hence larger distance d

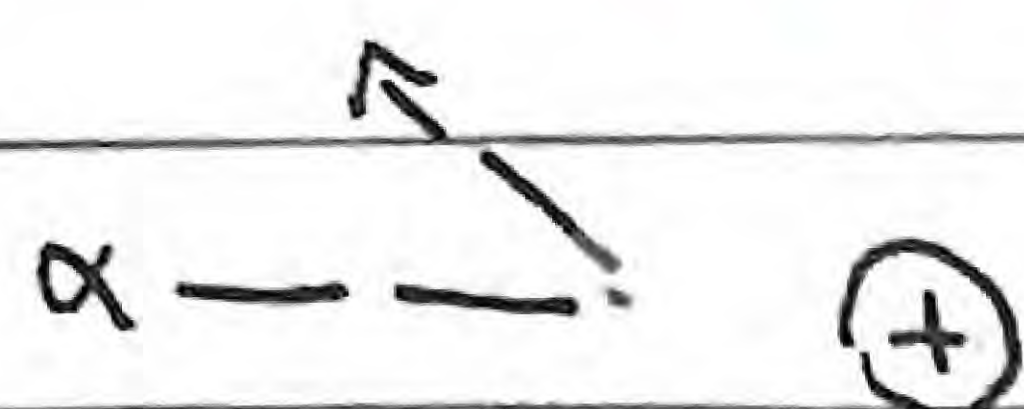
at higher E_k , α goes closer to nucleus (d decreases)

But at some point, there's no deflection

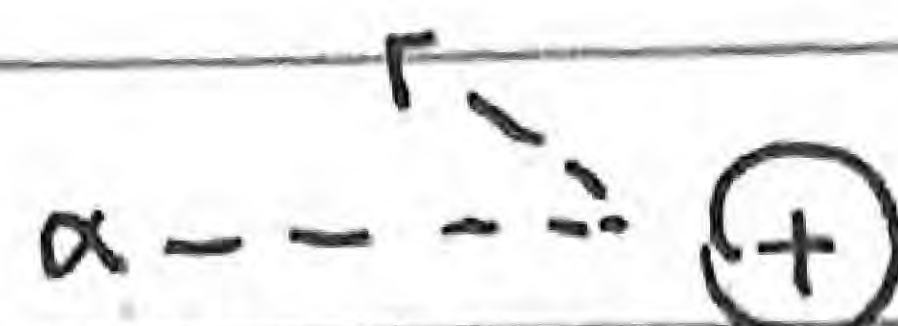
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↳ This is because lower distances mean strong nuclear force starts acting, overcomes repulsive force.

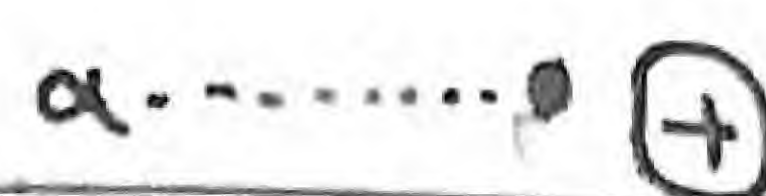
Size of nucleus



low E_{ki}
deflects



higher E_{ki}
deflects



highest E_{ki}
no deflection
(gets close enough
so strong nuclear force
starts acting)

↳ E_{ki} converts to E_{pf}

Hence, at the first no deflection E_{ki} , all energy is converted into potential energy

$$E_{ki} = E_{pf} = k \frac{Q_1 Q_2}{r} \quad \leftarrow r \text{ is closest approach distance}$$

exists because
no physical contact
between α and nucleus

let $Q_2 = \text{charge nucleus}$
then $Q_2 = \text{proton \#}$



Measured E_{ki}, Q_1, Q_2

$$r = \frac{k Q_1 Q_2}{E_{ki}} \quad \leftarrow \text{make sure } E_{ki} \text{ converted to joules}$$

↳ approximates radius of nucleus

This can help us calculate density...

$$\rho = \frac{m}{V} \quad \leftarrow m = A \cdot u \quad \leftarrow V = \frac{4}{3} \pi r^3 \quad \leftarrow \text{size of nucleus}$$

if given mass info

The size of nucleus when generalized is...

Fermi
radius

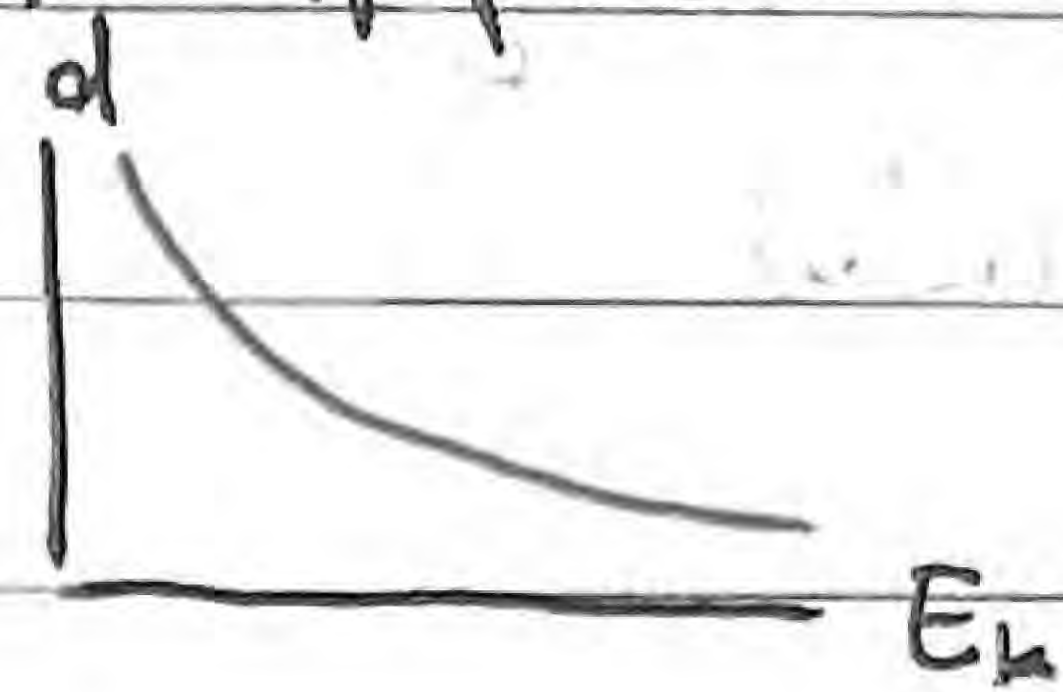
$$r = R_0 A^{1/3} \quad \leftarrow \text{nuclear radius}$$

for $R_0 = 7.24 \times 10^{-15} \text{ m}$, $A = \text{atomic mass}$

Size of nucleus

Graphs for distance of closest approach

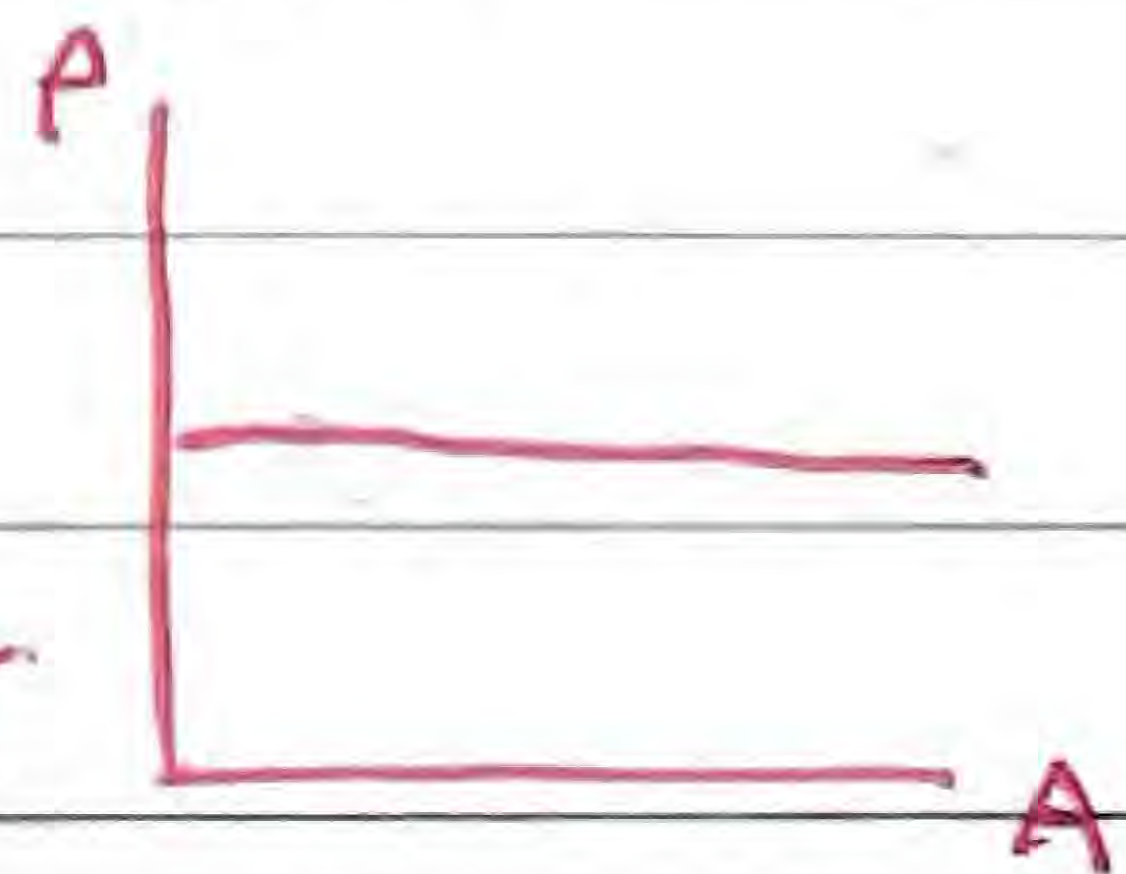
$$d = \frac{k q_1 q_2}{E_k} \quad \text{hence } d \propto \frac{1}{E_k}$$



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Nuclear Density can hence be simplified

$$\rho = \frac{m}{V} = \frac{A \cdot u}{\frac{4}{3}\pi R^3} = \frac{A \cdot u}{\frac{4}{3}\pi (R_0 A^{\frac{1}{3}})^3} = \frac{3u}{4\pi R_0^3}$$



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Note u and R_0 are constants

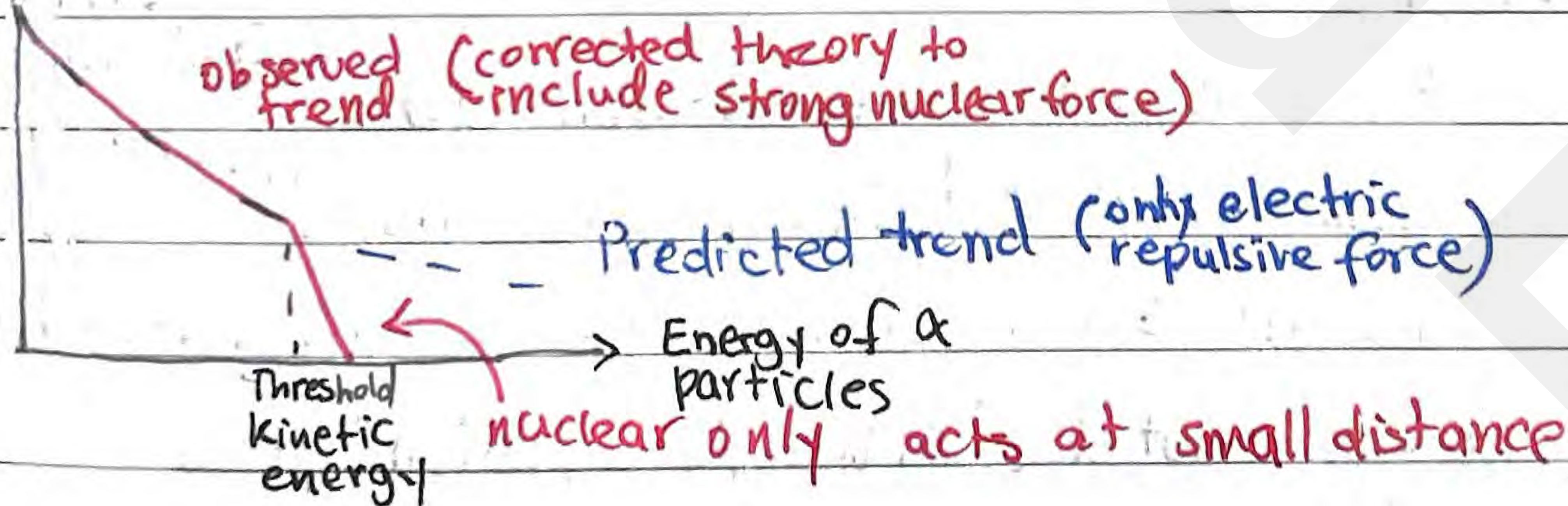
Hence $\rho \propto A$ (ρ only dependent on moles)

↳ All elements of same mole has same density

High Energy Rutherford Scattering

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↑ # of scattered α particles



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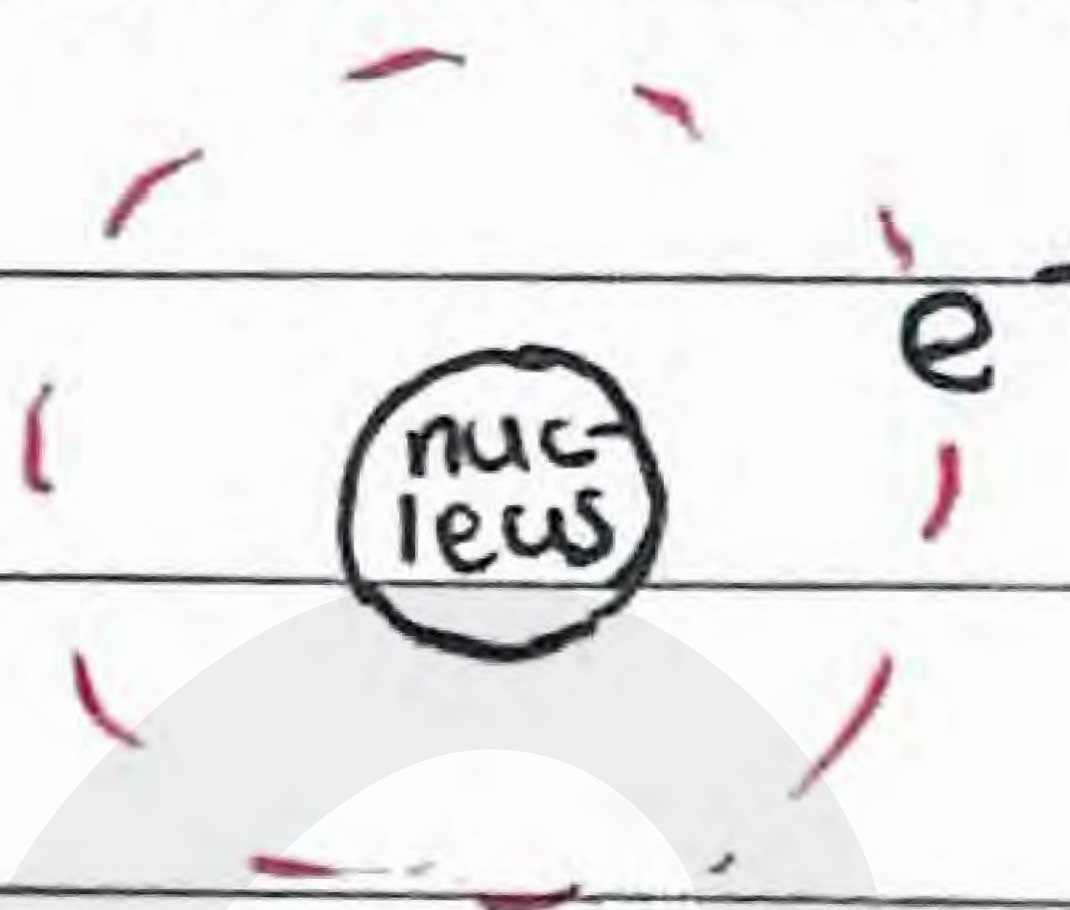
This deviation led to discovery of nuclear force

But Rutherford's model has a lot of issues

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Issues with Rutherford Model

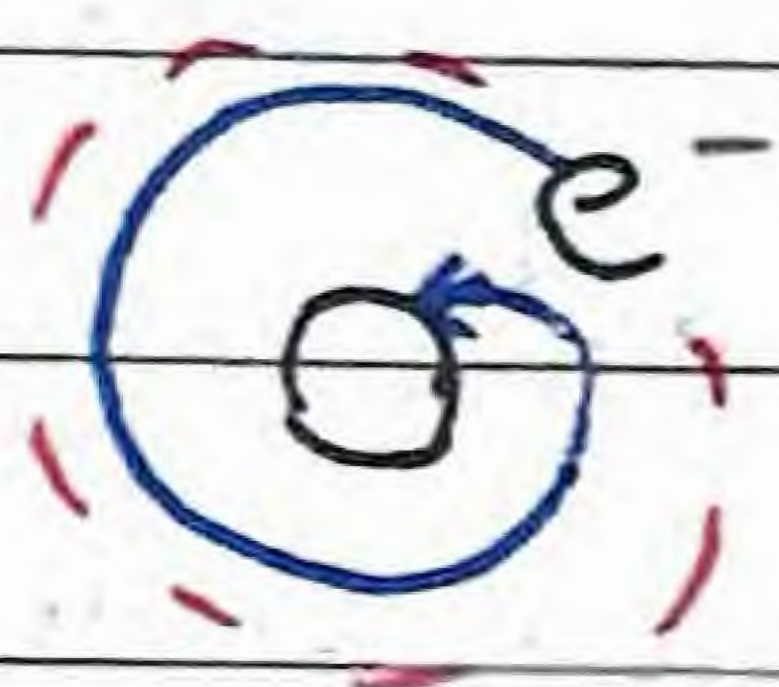
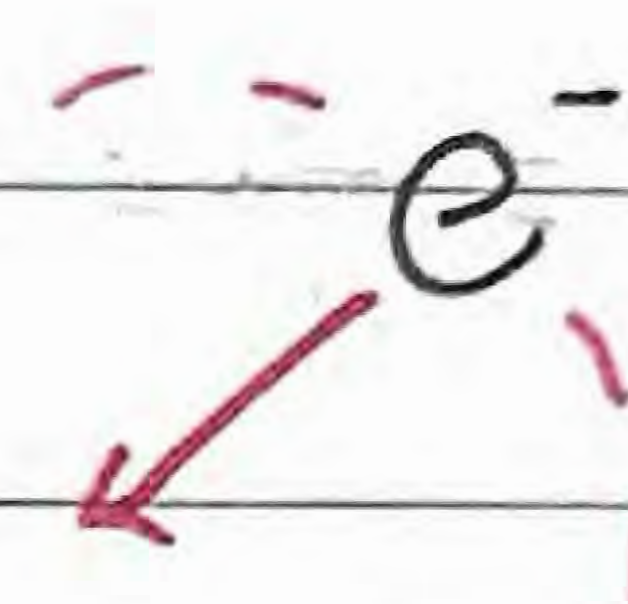
Rutherford's model assumes circular motion/orbit for electrons



However, with Electromagnetism Theory, anything accelerated must radiate energy

Maxwell's equation

Electrons are centripetally accelerated, hence must radiate and lose energy



But that means e^- can't maintain its orbit, thus a contradiction

So Bohr's model was introduced to fix this

Assumes ... 1. Discrete Orbits

2. Electrons located on fixed-orbits and cannot be located in between orbits

3. Transitions between orbits are discrete, non-continuous

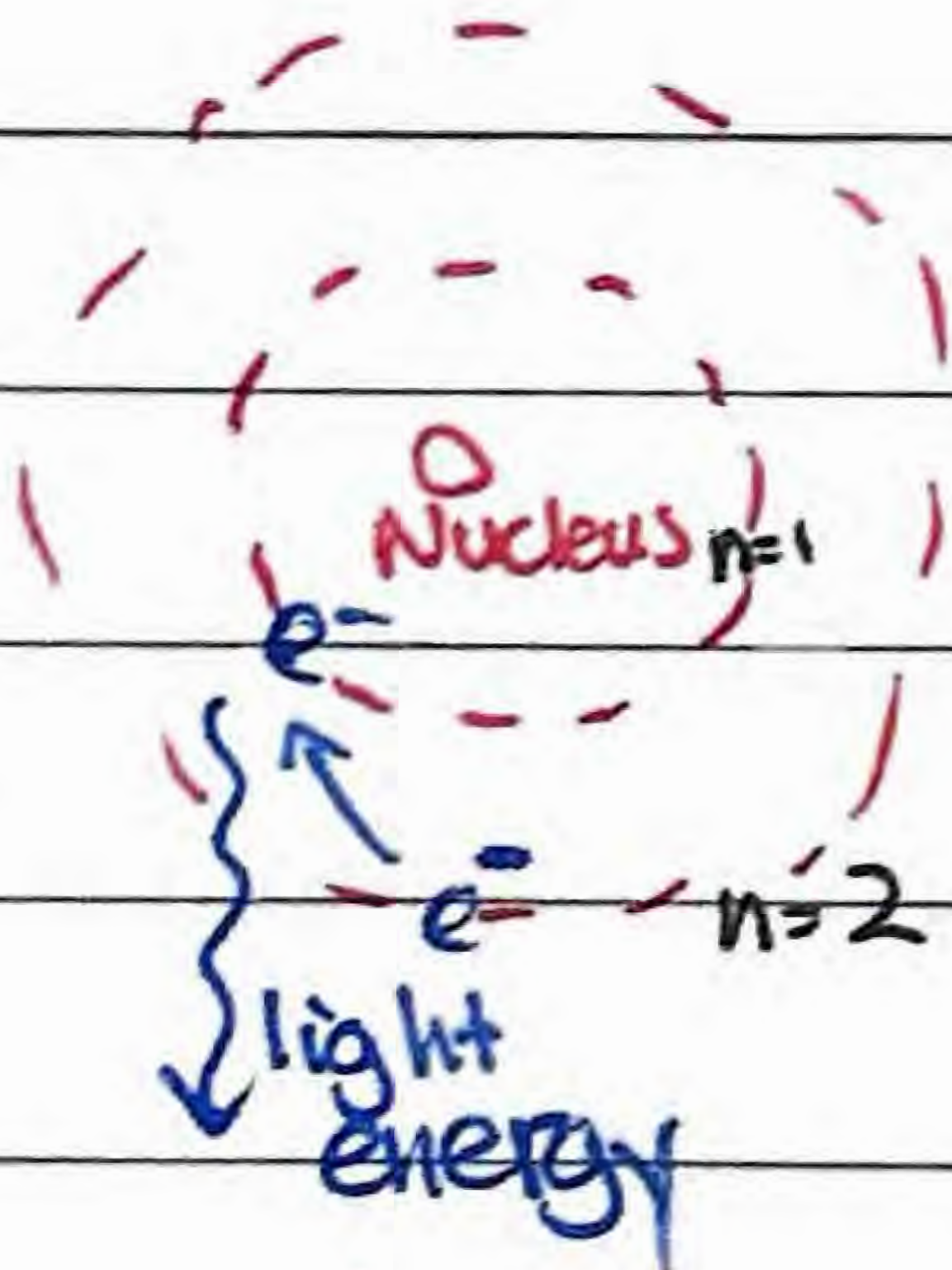
4. Does not use "Maxwell's Equation" for energy

5. Energy released interpreted as light: $\Delta E = h \cdot f$

$$\hookrightarrow \Delta E = \frac{hc}{\lambda} \Rightarrow \lambda = \frac{hc}{\Delta E}$$

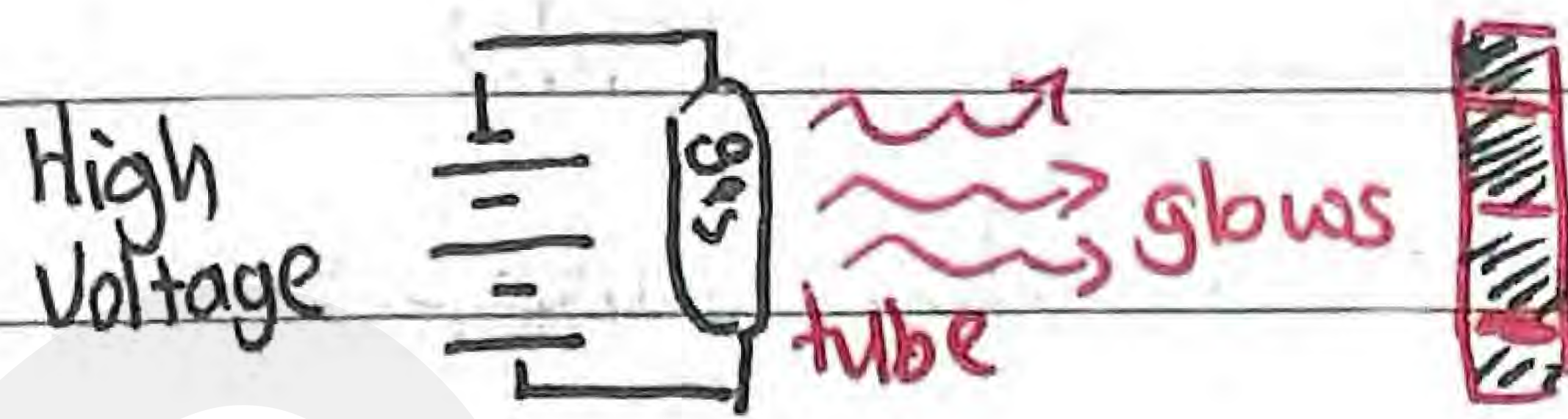
The λ s also compose the unique "fingerprint" of an element
 $\hookrightarrow$ verified by emission spectra experiment from 1890s

instantaneous
 ("teleports")



Emission Spectra

Bohr's theory aligns with Emission Spectra experiment



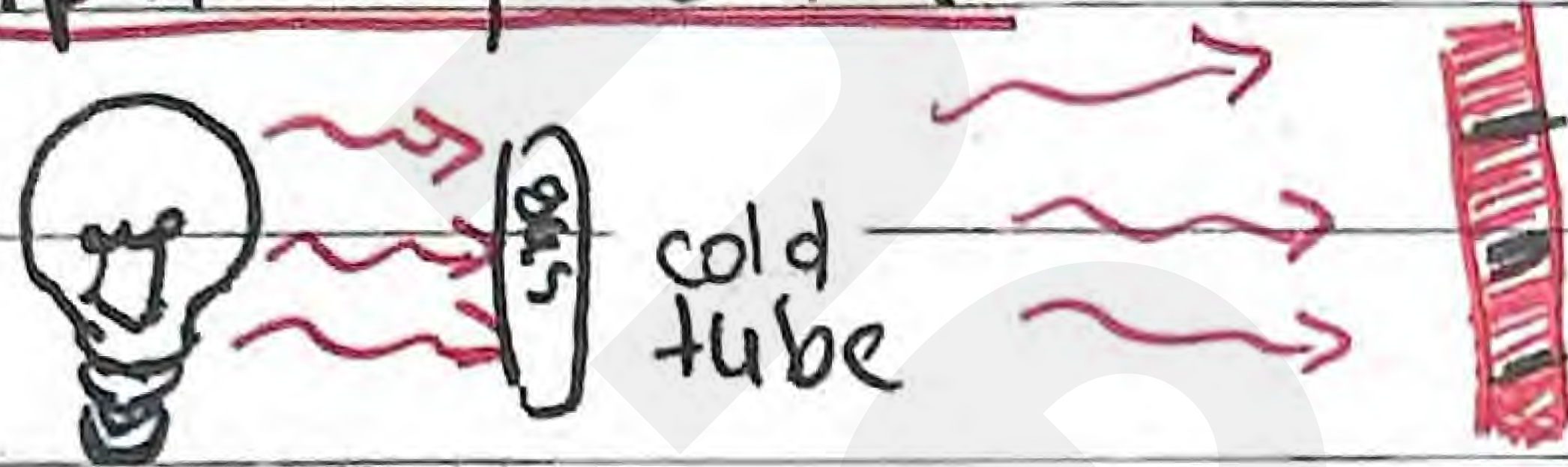
Hot gas

↳ Spectrum of light emitted by an element

• streaks of bright lines on a dark strip

→ there are ∞ streaks of lines, just only few are visible

Absorption spectra



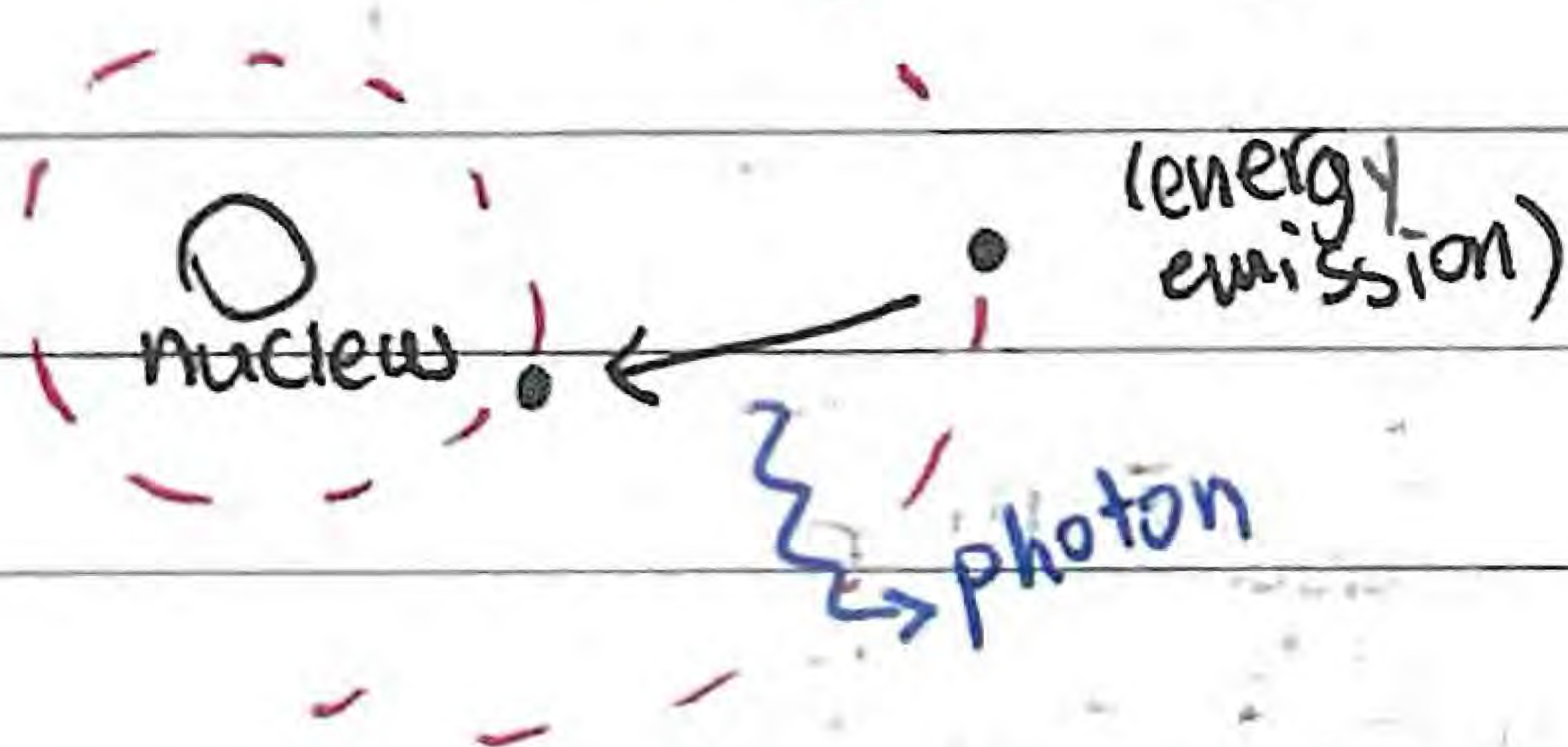
exact same λ s as

emission, except inverted

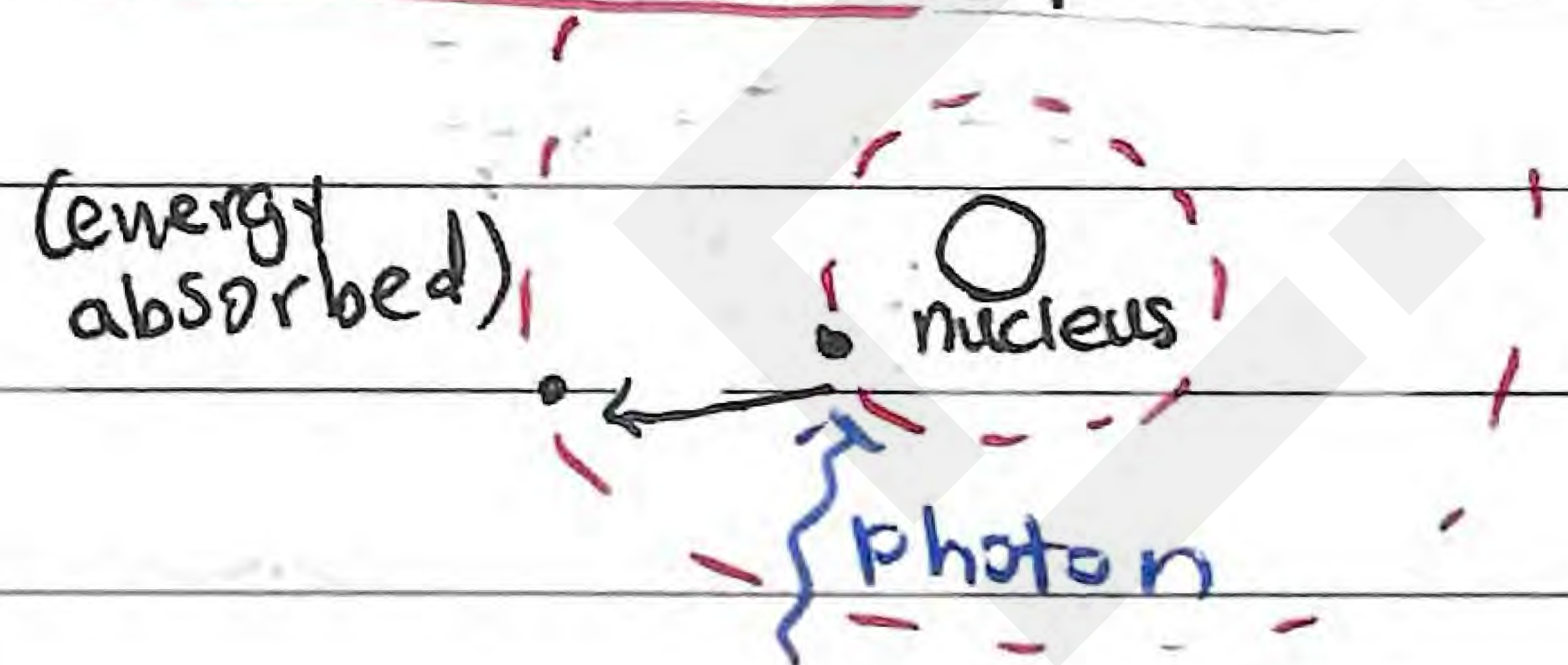
• Streaks of black lines on bright light/rainbow strip

→ there are ∞ streaks of lines, just only few are visible

Photon Emission (light output)



Photon Absorption (light input)



Energy emission/absorption

$$E = hf = \frac{hc}{\lambda}$$

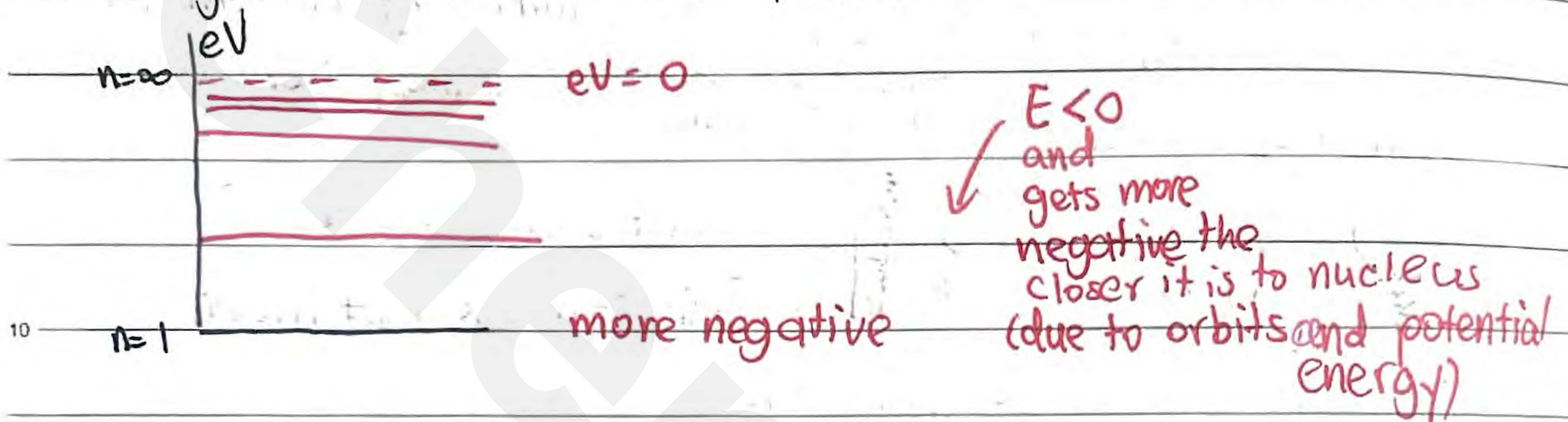
where h is planck's constant, $6.63 \times 10^{-34} \text{ J}\cdot\text{s}$

Electron Energy Levels

any level above is excited level. So 1st excited level is $n=2$

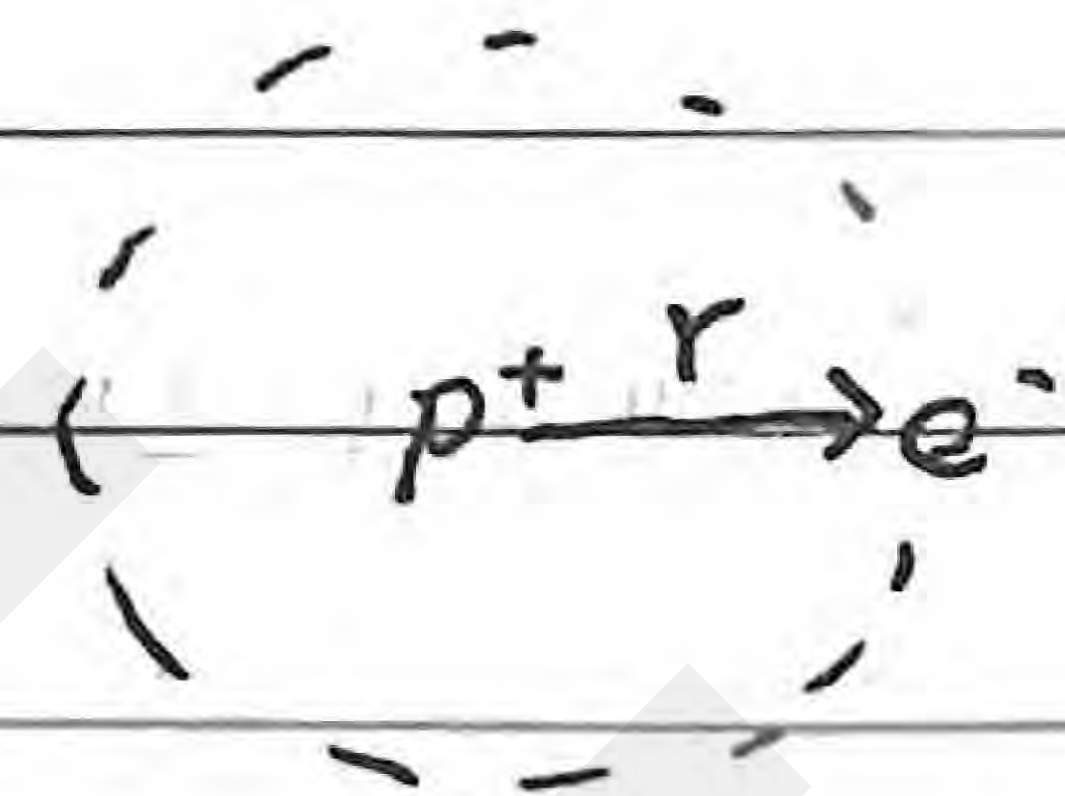
> $n=1$ level is always known as ground state

In questions with energy levels, energy is typically given but in eV, convert to Joules.



Quantization of angular momentum

Finding total energy of ^1H



$$\begin{aligned} E_T &= E_K + E_P \\ &= \frac{1}{2}mv^2 - k \frac{q_p q_e}{r} \\ &= \frac{1}{2}mv^2 - k \frac{e^2}{r} \\ &= -\frac{1}{2}k \frac{e^2}{r} \end{aligned}$$

classical physics

$$\begin{aligned} F_{\text{elec}} &= m \frac{v^2}{r} \\ k \frac{e \cdot e}{r^2} &= m \frac{v^2}{r} \\ mv^2 &= k \frac{e^2}{r} \end{aligned}$$

This makes sense because $-2E_K = E_P = 2E_T$ for bound orbital motion (recall gravitational orbital motion)

Proof for discrete energy levels...

Initially, Bohr hypothesized quantization of angular momentum

Böhr's condition

Hypothesis that angular momentum ($m \omega$)

$$mvr = \frac{nh}{2\pi} \quad \text{for } n \in \mathbb{Z}^+ (1, 2, 3, \dots)$$

↳ representing energy level

$$\therefore m^2 v^2 r^2 = \frac{n^2 h^2}{4\pi^2} \Rightarrow mv^2 = \frac{n^2 h^2}{4\pi^2 m r^2}$$

$m = \text{mass } e^-$

$$k \frac{e^2}{r} = \frac{n^2 h^2}{4\pi^2 m r^2}$$

Solving for radius of orbit $\approx 0.5 \times 10^{-10}$

$$r = \frac{h^2}{4\pi^2 k e^2 m} \cdot n^2 = (\text{constant}) n^2$$

for $n \in \mathbb{Z}^+$

Hence, from quantized angular momentum, radius of orbit is also quantized

when plugged into $E_T \dots \approx \frac{-13.6 \text{ eV}}{n^2}$

$$E_T = -\frac{1}{2} \cdot \frac{ke^2}{(\text{const})} \cdot \frac{1}{n^2} \approx -(\text{const}_2) \cdot \frac{1}{n^2}$$

↳ $\approx \frac{-13.6 \text{ eV}}{n^2}$ gives total energy for n -th energy level

Hence $E_{T\infty} = 0$

Since $E_T \propto \frac{1}{n^2}$, also dependent on an integer n , Böhr proved energy is quantized (discrete energy levels)

But where does Böhr's condition come from?

Einstein's Energy

$$E^2 = m^2 c^2 + p^2 c^2$$

$\neq 0$ for relativistic particles

For a photon (massless)

$$E^2 = p^2 c^2$$

← wait! but why is $p \neq 0$ if $p = mv$?

$$p = \frac{E}{c} = \frac{hf}{c}$$

notice formulas reveal $p = \frac{hf}{c} \neq 0$

$$\therefore p = \frac{h}{\lambda}$$

Momentum of a photon

"de Broglie's" formula

Rearranging...

$$\lambda = h/p = h/mv$$

classical mechanics

also works for quantum, just not photon

What does this mean?

↳ Anything with mass has wavelength

↳ Hence, any mass is also a wave

even a tennis ball is a wave

↳ classical mechanics - observed from particles
↳ modern physics - Also observed from waves

$\lambda = h/p$ reveal half of Bohr's reasoning, to further his background he thought of Electron-in-a-box model

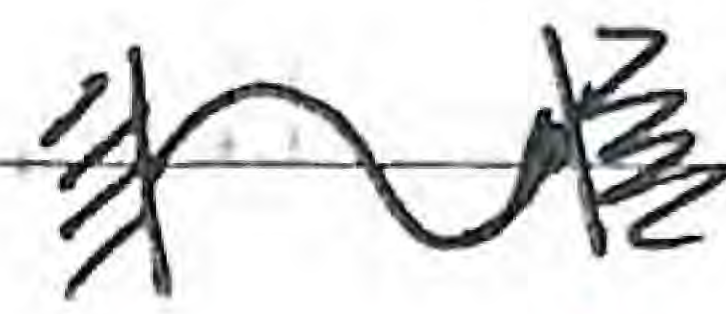
Electron-in-a-Box Model



makes sense
because of
confined orbit



alluding to



standing
waves
(harmonics)

Essentially, electrons
treated as standing waves

$$\lambda = \frac{2L}{n} \Leftrightarrow \frac{n}{2}\lambda = L$$

Combining with $\lambda = h/p \dots$

$$\therefore \frac{2L}{n} = \frac{h}{mv} \Rightarrow mv = \frac{hn}{2L}$$

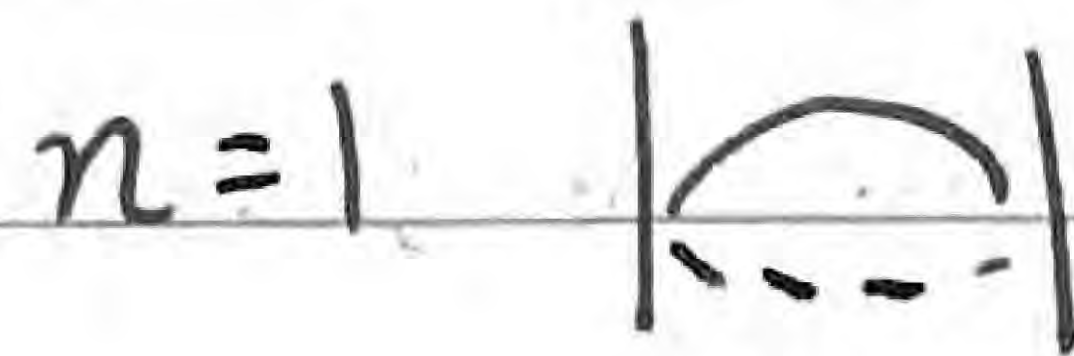
or... $\boxed{mvr = \frac{hnr}{2\pi}}$ Bohr's Condition

$$2\pi r = n \frac{h}{mv} = n\lambda$$

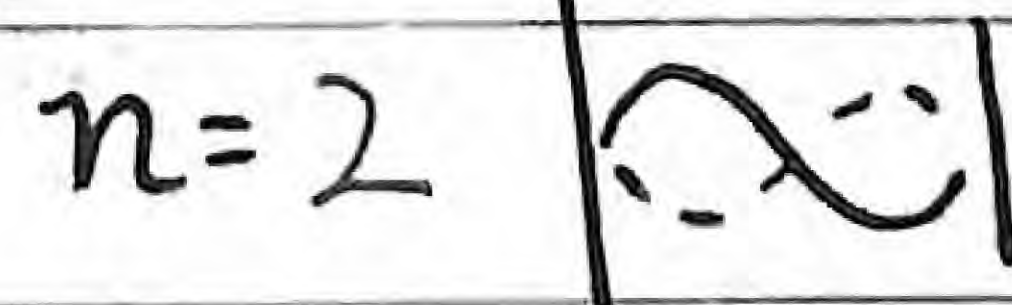
Hence... $2\pi r = \text{circumference} = n\lambda$

...suggesting a circumference and thus
radius varying with n

Recall standing waves...



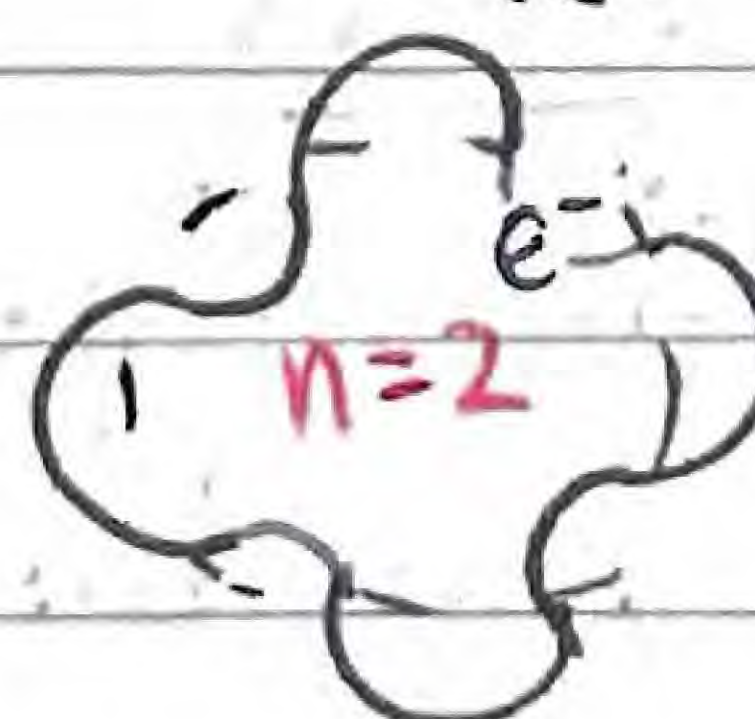
(several loops)



etc.

but you
orbit in
a loop

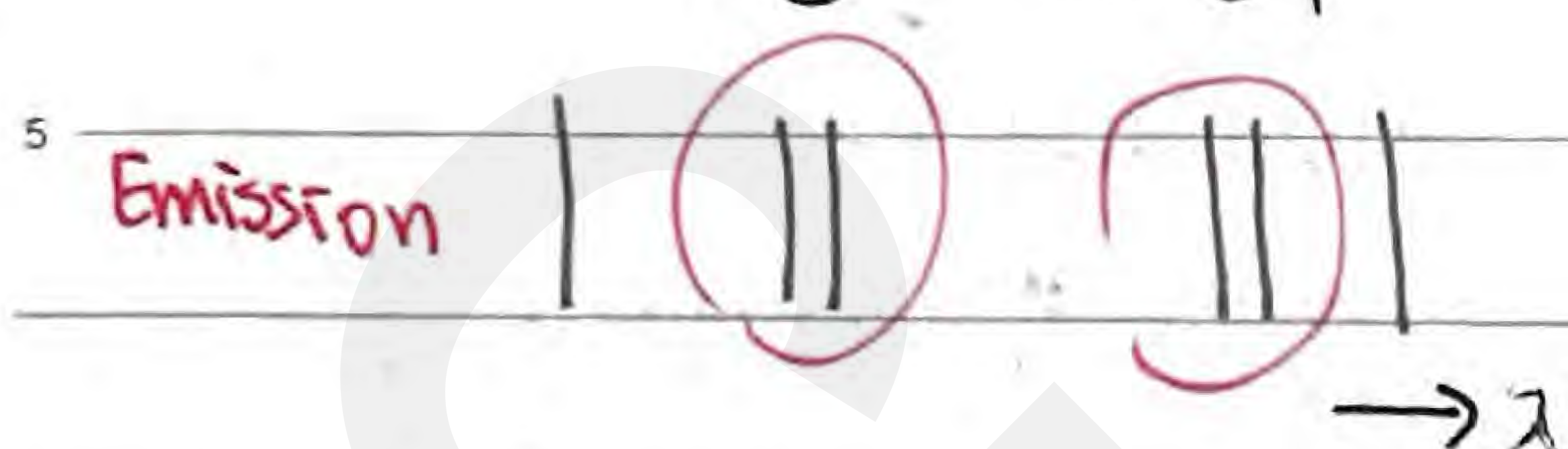
Thus, electron orbits are like



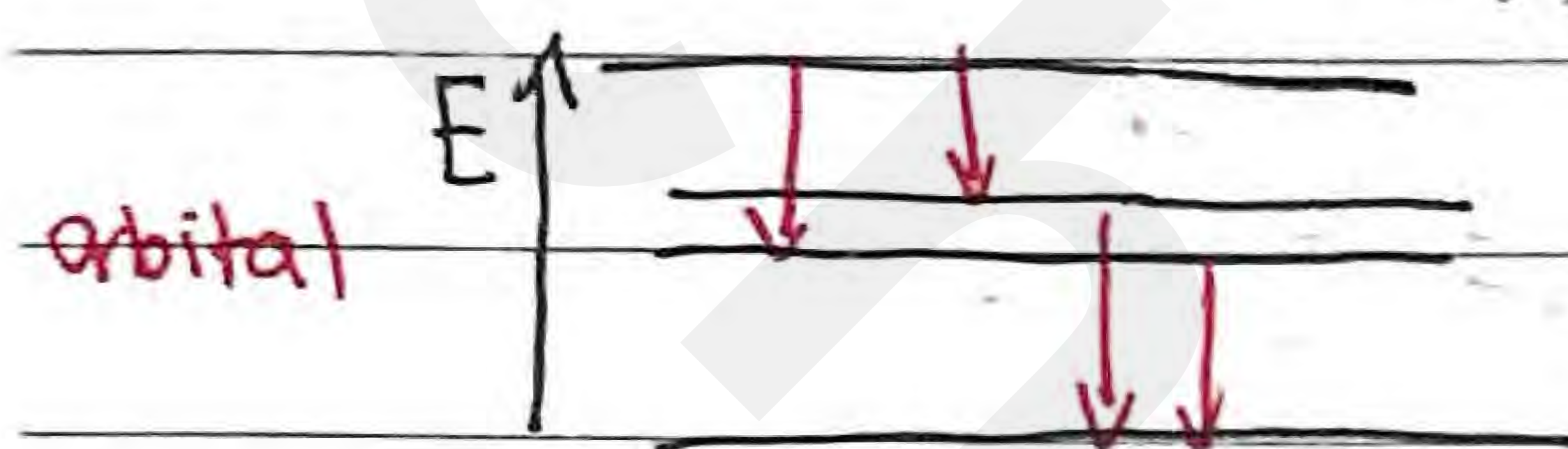
$\Rightarrow$ electron
orbitals!

Emission Spectrum

Matching energy levels to emission spectrum



consider # of combinations
 $\binom{n}{2}$



some streaks close together

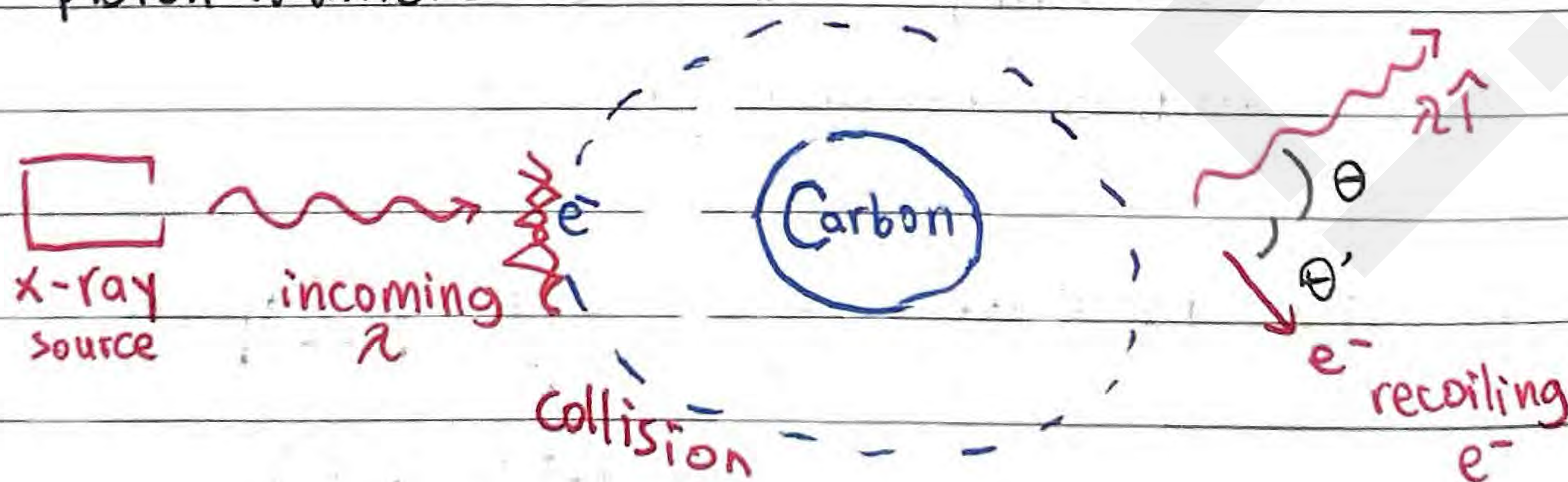
→ And closeness of energy transitions

→ so transitions must be of similar energy changes

The Compton Effect

In 1923, Compton discovered that the wavelength of x-rays increased after scattering

→ The x-rays were scattered by elements of small proton numbers



before scattering

after scattering

x-ray collides with e^- of carbon, wave (λ') scattered at an angle and hits e^- off of carbon

→ energy of photon transferred to e^-

Compton Equation

Takes into account energy and momentum conservation

$$\rightarrow \Delta\lambda = \frac{h}{m_e c} (1 - \cos \theta) = \lambda' - \lambda$$

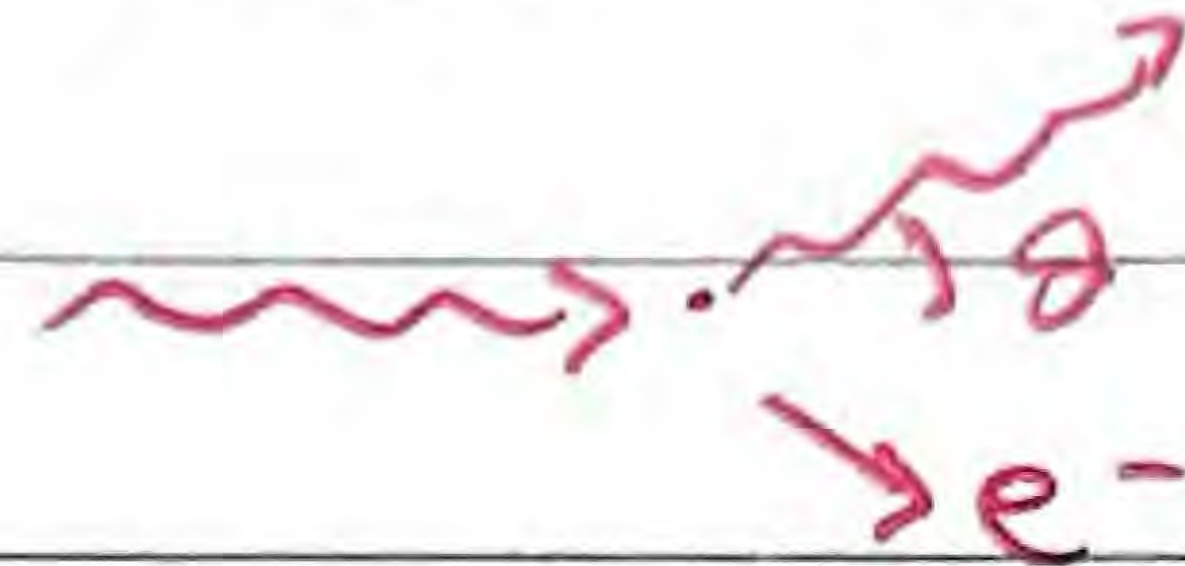
if $\theta = 0$, no scattering
↓
so no change in

where ...

λ' - Final wavelength, λ - Initial wavelength

m_e - mass electron, h - Planck's Constant

θ - light scatter angle, c - light speed



The initial wavelength can only be x-ray or gamma because only those have sufficient energy

$\rightarrow \lambda \uparrow$ because some energy of photon is transferred to electron

Compton scattering of light is evidence of the particle nature of light

Conservation of energy... $E_{\text{recoil } e^-} = E_{\text{lost by photon}}$

or... $E_{\text{photon, init}} = E_{\text{photon, final}} + E_{e^-}$